

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient backend workflow for MCA filings with retries, session recovery, public-site API lookups, and state validation, reducing manual filing time by **85%**.
- Optimized an AI-assisted company-registration pipeline for name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across workflow preparation.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling preprocessing, orientation correction, model inference, generated masks, and large-volume data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged complex pipeline issues across conformation, orientation, label mapping, PyTorch/ONNX-style inference, and binary mask generation in a mature open-source codebase.

Open Source

[Contributions](#) 

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Seaweed Contest Platform | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

[GitHub](#) 

- Built backend services for registration, code submissions, sandboxed Judge0 execution, scoring, and live rankings across **500+ concurrent users**.
- Designed async dispatch, verdict polling, PostgreSQL result schemas, Redis-backed runtime state, and Docker/Kubernetes deployment on AWS, reaching **sub-5s P95 verdict latency**.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

[GitHub](#) 

- Built Go/Echo backend services for a content distribution platform with REST APIs, RabbitMQ workers, Redis autosave, PostgreSQL persistence, and recoverable job state.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so unreliable external API/browser workflows stayed observable and recoverable.

Swish Support System | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built operational backend workflows for order status, delays, refunds, trust checks, and escalation using Python/Go services with PostgreSQL and Redis state.
- Maintained per-conversation order context, evidence artifacts, deterministic policy decisions, and traceable action records across multi-turn customer-support flows.

Technical Skills

Languages: Python, Go, TypeScript, C++

Frameworks: FastAPI, Gin, Echo, Node.js

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, AWS, GCP, GitHub Actions, Prometheus, OpenTelemetry